

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	
Team ID	
Project Name	Virtual Agent–Driven SLA Breach Awareness & Justification System
Maximum Marks	4 Marks

Technical Architecture

The Virtual Agent–Driven SLA Breach Awareness & Justification System is built on the ServiceNow platform using a low-code architecture. The system integrates Incident Management, SLA monitoring, Flow Designer, Virtual Agent, Email Notifications, Platform Analytics, and Reporting to automate SLA governance. When an incident approaches an SLA breach, automated workflows notify users, collect mandatory justifications through the Virtual Agent, update incident records, and provide managers with real-time dashboards and reports for monitoring and decision-making.

Table-1: Technology Stack (Architecture & Stack)

S.No	Component	Description	Technology
1	User Interface	Interface for IT Support Engineers, Managers and Administrators	ServiceNow Workspace, Platform UI
2	Application Logic-1	Incident and SLA monitoring	ServiceNow Business Rules
3	Application Logic-2	Workflow automation for SLA breach handling	Flow Designer
4	Application Logic-3	User interaction and justification collection	Virtual Agent
5	Database	Stores incidents, SLA records, breach reasons and justifications	ServiceNow Database
6	Cloud Platform	Cloud-hosted application environment	ServiceNow Cloud Platform
7	Reporting & Analytics	SLA dashboards, charts and performance reports	Platform Analytics, Reports & Dashboards
8	Notification Service	Sends email alerts and manager notifications	Notification Engine, Email Action

9	Integration API	Sends notifications and workflow events	ServiceNow REST APIs
10	AI Component	Conversational assistance for SLA acknowledgement	ServiceNow Virtual Agent
11	Infrastructure	Application hosting and deployment	ServiceNow SaaS Cloud Infrastructure

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Low-code platform with configurable workflows	ServiceNow Platform
2	Security Implementations	Role-based access, authentication, audit logs and data protection	ACLs, Roles, User Authentication
3	Scalable Architecture	Supports multiple incidents, users and SLA records simultaneously	ServiceNow Cloud Architecture
4	Availability	Provides continuous monitoring and high system uptime	ServiceNow Cloud Infrastructure
5	Performance	Real-time SLA monitoring, automated notifications and dashboard updates	Flow Designer, Platform Analytics